Parul® University
Vadodara, Gujarat

NAAC GRADE A++

Information and Communication Technology

Exception Handling

Study Guide

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8.2.1 Exception Handling using five keywords

✓ 1. try Keyword

➤ **Definition:**
The try block in Java is used to wrap code that might throw an exception during execution.
➤ **Syntax:**

```
try {
   // Risky code that may throw an exception
}
```

➤ **Purpose:**
- Prevents the abrupt termination of the program.
- Is always followed by one or more catch blocks or a finally block.

➤ **Example:**

```
try {
   int num = 10 / 0;
}
```

✓ 2. catch Keyword

➤ **Definition:**
The catch block handles the exception thrown by the try block. It defines what should be done if a specific type of exception occurs.
➤ **Syntax:**

```
try {
   // Risky code
} catch (ExceptionType e) {
   // Handling code
}
```

➤ **Multiple catch blocks:**
You can have multiple catch blocks for handling different exceptions:

```
try {
   int[] arr = new int[5];
   arr[10] = 3;
} catch (ArrayIndexOutOfBoundsException e) {
   System.out.println("Array index issue");
```

```
} catch (Exception e) {
    System.out.println("General exception");
}
```

🔷 Flow of Control in Exception Handling

- When an exception occurs inside the try block:
    1. The control transfers to the **first matching catch block**.
    2. If no catch block matches, the exception is propagated up the call stack.
    3. Regardless of what happens, the **finally block** (if present) is executed.

✅ 3. finally Keyword

➤ **Definition:**
The finally block is always executed after the try and catch blocks, regardless of whether an exception was thrown or not.

➤ **Syntax:**

```
try {
    // Risky code
} catch (Exception e) {
    // Handle exception
} finally {
    // Cleanup code
}
```

➤ **Use Case:**

- Closing resources like files, database connections, etc.

➤ **Example:**

```
try {
    System.out.println("Try block");
} catch (Exception e) {
    System.out.println("Catch block");
} finally {
    System.out.println("Finally block always runs");
}
```

✅ 4. throw Keyword

➤ **Definition:**
The throw keyword is used to **explicitly throw an exception**.

➤ **Syntax:**

```
throw new ExceptionType("Error Message");
```

➤ **Rules:**

- Used inside method body.
- Can throw only one exception at a time.
- The thrown exception must be of type Throwable.

➤ **Example:**

```
public class Test {
    public static void main(String[] args) {
        throw new ArithmeticException("Manually thrown exception");
    }
}
```

✅ 5. throws Keyword

➤ **Definition:**

The throws keyword is used in a method signature to **declare the exceptions** that the method might throw.

➤ **Syntax:**

```
void myMethod() throws IOException, SQLException {
    // Method code
}
```

➤ **Use Case:**

- To inform the caller method about possible exceptions.
- Often used with checked exceptions.

➤ **Example:**

```
import java.io.*;

public class Test {
    void readFile() throws IOException {
        FileReader fr = new FileReader("data.txt");
    }
}
```

➢ Summary of Key Concepts:

- **Java uses try-catch-finally blocks for handling exceptions.**
- **Multiple and nested catch blocks allow granular control.**
- **throw is used to raise exceptions manually.**
- **throws indicates what exceptions a method may propagate.**

➢References:

- **Oracle Java Documentation – Exception Handling**
  🔗 https://docs.oracle.com/javase/tutorial/essential/exceptions/
- **GeeksforGeeks – try, catch, throw and throws in Java**
  🔗 https://www.geeksforgeeks.org/exceptions-in-java/
- **Baeldung – Java Exception Handling**
  🔗 https://www.baeldung.com/java-exceptions
- **JavaTpoint – Java Exception Handling**
  🔗 https://www.javatpoint.com/exception-handling-in-java

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